MIL-I-43728

31 December 1970
See 6.6

MILITARY SPECIFICATION

INJECTOR, RINSE, AUTOMATIC, FOR COMMERCIAL DISHWASHING MACHINES

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a unit for automatically injecting a liquid drying agent into the final rinse water of commercial dishwashing machines.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

P-R-1272	-	Rinse, Additive, Dishwashing
PPP-B-636	-	Box, Fiberboard
PPP-B-566	-	Box, Folding, Paperboard
PPP-B-676	-	Box, Setup

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 7320

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2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to American Trucking Associations, ATTN: Tariff Order Section, 1616 P Street, N.W., Washington, D. C. 20036.)

NATIONAL SANITATION FOUNDATION (NSF)

Listing of Food Service Equipment

Standard No. 29 - Detergent and Chemical Feeders for Commercial Spray Type Dishwashing Machines.

(Application for copies should be addressed to the National Sanitation Foundation, P.O. Box 1468, Ann Arbor, Michigan 48106.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 202 Union Station, 516 W. Jackson Blvd., Chicago, Illinois 60606.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS(ASTM)

Standard A167 - Corrosion-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip.

Standard A176 - Corrosion-Resisting Chromium Steel Plate, Sheet, and Strip.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

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UNDERWRITERS' LABORATORIES, INC. (UL)

Standard 921 - Commercial Electric Dishwashers

(Application for copies should be addressed to the Underwriters' Laboratories Inc., 207 East Ohio Street, Chicago, Illinois 60611; 1284 Walt Whitman Road, Melville, Long Island, New York 11746; or 1655 Scott Boulevard, Santa Clara, California 95050.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article.- When specified (see 6.2), the contractor shall furnish a sample for first article inspection and approval (see 4.2 and 6.3).

3.2 Standard product.- The rinse injector delivered under this specification shall be the manufacturer's current standard commercial product, except for any changes necessary to comply with specification requirements.

3.3 Standards.-

3.3.1 UL standards.- The rinse injector shall conform to the applicable requirements of UL Standard 921.

3.3.1.1 UL certification.- Prior to approval of the preproduction sample, if one is submitted, or prior to approval of the first shipment, the supplier shall submit satisfactory evidence to the contracting officer or his authorized representative that the rinse injectors he proposes to furnish under this specification meet the applicable requirements of UL Standard 921. Acceptable evidence of meeting the applicable requirements of UL standards shall be the UL label, or listing mark, or a certified test report from a recognized independent testing laboratory acceptable to the Government indicating that the rinse injectors offered have been tested and conform to the requirements of UL standards as applicable.

3.3.2 NSF standard.- Rinse injectors shall conform to the applicable requirements of NSF Standard No. 29.

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3.? 2.1 NSF certification.- The supplier shall submit satisfactory evidence to th? contracting officer or his authorized representative that the rinse injec ors he proposes to furnish under this specification meets the applicable requi ements of NSF Standard No. 29 as follows:

1) Listing in the current edition of the NSF "Listing of Food Service Equipment" or display of the NSF seal on the finished rinse injector shall be submitted prior to approval of the preproduction sample, if one is submitted, or prior to proval of the first shipment, or

2) A ertificaton for the rinse injector issued by NSF under their speci.l one-time evaluation/certification service shall be submitted prior to approval of the preproduction sample, if one is submitted, or prior to approval of the first shipment, or

(3) A certified test report from a recognized independent testing laboratory acceptable to the medical department of the service for which the injectors are being procured, indicating that the rinse injector has been tested and conforms to NSF Standard No. 29 shall be submitted prior to the approval of the preproduction sample, if one is submitted, or prior to commencing production.

3.? Design.- The rinse injector shall be designed for use with rinse agent conforming to P-R-1272 to automatically inject a liquid drying agent into the final rinse water of commercial dishwashing machines to accomplish rapid sheeting and drying for the purpose of eliminating streaks and droplet on tableware and flatware. The injector shall operate on rinse water line flow or be pressure-activated and shall maintain a preset concentration of rinse agent (see 6.4) in the final rinse water within a tolerance of 5 percent over a ıge of water flow of 4 to 15 gallons per minute or 15 to 30 pounds per square inch water pressure regardless of fluctuations in flow or pressure (see 4.4.1). The injector shall be for installation directly in the final rinse water line or mounted on the dishwashing machine body and shall inject the rinse agent into the downstream side of the rinse valve. Reservoirs shall be for mounting on the machine or on an adjacent wall. All necessary tubing and wiring to form a complete system shall be provided as specified (see 6.2).

3.5 Materials.- Materials not definitely specified shall be of the quality normally used by the manufacturer for rinse injectors provided the completed item complies with all provisions of this specification.

3.5.1 Corrosion-resisting steel.- Corrosion-resisting steel shall conform to the 300 or 400 series as specified respectively in ASTM Standard A167 or ASTM Standard A176.

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3.6 Construction.-

3.6.1 Reservoir.- The drying agent shall be contained in a reservoir and shall be installed so that operator can visually determine liquid level. The reservoir shall be corrosion-resisting steel specified in 3.5.1 or polyethylene. Capacity shall be not less than 32 fluid ounces. A corrosion-resisting steel bracket for attaching units to the machine or wall shall be provided.

3.6.2 Adjustment.- Injectors shall have a manual adjustment for obtaining desired proportion of rinse agent in the final rinse water line.

3.6.3 Electrical characteristics.- Unless otherwise specified (see 6.2), the electrically operated injectors shall be designed for operation on a nominal 120 volt, 60 hertz, 1-phase electrical system.

3.7 Marking for identification.- The manufacturer's name, contract number model number, and serial number shall be permanently marked on a corrosion-resistant plate and affixed to the injector.

3.8 Finish.- The injector components shall be finished in accordance with the manufacturer's standard commercial practice.

3.9 Operating instructions.- Instructions indicating the proper operating procedures for using the rinse injector shall be etched, stamped or embossed on a brass plate or corrosion-resistant metal or plastic plate and permanently attached in a visible position for the operator.

3.10 Workmanship.- Components shall be finished with all burrs and rough edges removed. Welds shall be sound, smooth, and free from pits, fractures, burn-through, or fissures. Spot welds shall be free from pits and flashes. Bolt holes and rivet holes shall be accurately punched or drilled and shall have burrs removed. Washers or lockwashers shall be provided in accordance with standard commercial practice, and all bolts, nuts, and screws shall be drawn tight. Rivets shall be tightly driven, fully peened, and shall completely fill the holes. Electrical wiring shall be provided with adequate slack to relieve strain and the wiring shall not be abraded, poorly joined, or loose at terminals.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless othewise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in

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6th Ordinance Br

the contract or order, the supplier may use his own or any other facilities suitable for performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

.2 First article inspection.- When a preproduction sample is required, it shall be examined for defects listed in table I and tested as specified in 4.4.1. Failure to pass any tests shall be cause for rejection of the preproduction sample. The presence of any defects listed in table I or any dimension not within specified requirements shall be cause for rejection of the preproduction sample.

4.3 Inspection.- Sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3.2 End item inspection.- The inspection lot shall consist of all injectors offered for inspection at one time. The sample unit shall be one completely fabricated injector.

4.3.2.1 Visual examination.- The injector shall be examined for defects in table I. The inspection shall be level II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE I.- Classification of defects

Examination	Defect	Classification Major	Minor
Construction and workmanship	Any component missing or not as specified	X	
	Any component fractured or malformed	X	
	Any burr or rough edge	X	

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TABLE I.- Classification of defects (cont'd)

Examination	Defect	Classification Major	Minor
Construction and workmanship (cont'd)	Functioning component that requires abnormal force to operate	X	
	Any adjustable assembly not adjusted to perform the function intended		X
	Any component not accessible for replacement or repair	X	
Welding	Any weld fractured, not sound, or contains fissures or burn-through	X	
	Any weld not smooth or contains pits		X
	Spot weld contains flashes		X
Bolts, nuts, screws	Stripped or fractured	X	
	Loose or not provided with lockwashers		X
Rivets	Loose, not fully peened, or fractured	X	
Electrical assemblies (when applicable)	Wiring cut, abraded, or loosely joined at terminals	X	
	Adequate slack not provided for wiring to relieve strain	X	
	Insulation stripped from wire	X	
Identification marking	Missing, incomplete, not legible, not located as specified		X

4.3.2.2 Capacity examination.- Capacity examination shall be made of the reservoir to determine compliance with this specification. Any reservoir capacity that is not within the specified requirements shall be classified as a defect. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of defects per hundred units.

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4.3.3 Testing of the end item.- When a preproduction sample is not required, the initial production unit shall be tested as specified in 4.4.1. In addition, one unit from each production lot shall be tested as specified in 4.4.1.

4.3.4 Standards compliance.- Proof of compliance with the applicable requirements of 3.3.1 and 3.3.2 shall be submitted to the Government representative.

4.4 Examination of preparation for delivery.- An examination shall be made to determine that packaging, packing, and marking as required by section 5 of this specification are complied with. Defects shall be scored as specified in table II. The sample unit shall be one shipping container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-4 with an AQL of 6.5 defects expressed in terms of defects per hundred units.

TABLE II.- Examination of preparation for delivery

Examination	Defect
Markings	Omitted; incorrect; illegible; of improper size; location, sequence, or method of application.
Materials	Components missing, damaged, or otherwise defective.

4.5.1 Examination of shipping containers.- When shipping containers are required to be in accordance with PPP-B-636, examination for defects in closure, waterproofing, and strapping shall be in accordance with PPP-B-636.

4.4 Test.-

4.4.1 Operation.- The injector shall be installed on a dishwashing machine in accordance with the manufacturer's installation instructions. Rinse agent used for the test shall conform to P-R-1272. A measured amount of rinse aid shall be added to the dispenser, the dispenser shall be adjusted to its lowest setting, and the dishwasher started. The final rinse shall be activated and maintained in an operating position so that the rinse aid injector is in constant operation. The water flow rate of the final rinse shall be varied in accordance with 3.4 and measured. The rinse aid injector shall be adjusted, by measuring the amount of rinse aid consumed compared to the original amount measured to provide a .01 percent concentration of rinse aid over the flow range specified in 3.4. When this setting has been achieved, the dishwasher and injector shall be allowed to operate for 15 minutes to determine if the dispenser will maintain this concentration within the tolerances specified in 3.4. This operation shall be repeated four times. Any failure of the injector to maintain the specified rinse concentration shall constitute failure of the test.

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5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.- The components of the injector shall be packaged in snug-fitting folding or set-up paperboard or fiberboard boxes conforming to variety 1, style and type optional of PPP-B-566, variety 1, type, class, and style optional of PPP-B-676, or type CF, variety SW, or type SF, class domestic of PPP-B-636, respectively, except that the weight of contents of paperboard boxes shall not exceed 10 pounds. Components shall be cushioned, as necessary, to prevent movement or damage while in transit. Closure shall be in accordance with the appendix of the applicable box specification.

5.1.2 Level C.- The injectors shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing.- Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.- Each complete injector packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, V2s of PPP-B-636. Closure, waterproofing by means of tape, and reinforcing with flat strapping or tape banding, shall be in accordance with the appendix of PPP-B-636.

5.2.2 Level B.- Each complete injector, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF, variety SW, or type SF, class domestic of PPP-B-636. Closure shall be in accordance with method II of the appendix of PPP-B-636.

5.2.2.1 When specified (see 6.2), the shipping container shall be a grade V3c, V3s or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix thereto.

5.2.3 Level C.- Injectors, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shalll be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking.- In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use.- The rinse injectors covered herein are intended for use on single and multiple tank commercial dishwashing machines.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) When a first article is required (see 3.1, 4.2, and 6.3).
(c) Location of injector (see 3.4).
(d) Electrical characteristics if other than specified (see 3.6.3).
(e) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(f) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).

6.3 First article.- When a first article is required, it shall be tested and approved under the appropriate provisions of paragraph 7-104.55 of the Armed Services Procurement Regulations. The first article should be a pre-production sample. The first article should consist of one unit. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examination, test and approval of the first article.

6.4 Liquid rinse additive for use with the rinse injector covered by this specification may be requisitioned from GSA by referencing Federal Stock Number (FSN) 7930-685-4762 (1/2-gallon) or FSN 7930-619-9575 (1-gallon).

6.5 Any requirements for equipment manuals for the item covered by this specification should be included in DD Form 1423 Contract Data Requirements List and cited in the contract.

6.6 This specification contains the requirements of Interim Purchase Description IP/DES S-40-7, dated 5 September 1967.

Custodians:

Army - GL
Navy - YD
Air Force - 84

Preparing activity:

Army - GL

Project No. 7320-0469

Review activities:

Army - MD
Navy - MC, MS

User activities:

Army - CE
Navy - SH

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-I-43728

Injector, Rinse, Automatic, For Commercial Dishwashing Machines

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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